2026/10/22 Business Intel.

(8) Single Linear Neuron.

Given

$$x = \begin{pmatrix} 2 \\ -1 \end{pmatrix}, \quad w = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad b = 0, \quad y = 3$$

a) Compute forward pass

$$\hat{y} = w^T x + b = w^T x$$

and the loss

$$\frac{1}{2}(\hat{y} - y)^2$$

$$\Rightarrow \hat{y} = \begin{pmatrix} 2 \\ -1 \end{pmatrix} \begin{pmatrix} 1 & 0 \end{pmatrix} = (2) \checkmark \checkmark$$

$$\Rightarrow L = \frac{1}{2}(2-3)^2 = \left(\frac{1}{2}\right) \checkmark$$

b) Compute the gradients

$$\frac{\partial L}{\partial w}, \quad \frac{\partial L}{\partial b}$$

$\Rightarrow$ We must express L in terms of w and b

$$\Rightarrow L = \frac{1}{2}(\hat{y} - y)^2 = \frac{1}{2}[(w^T x + b) - y]^2$$

Chain rule: $h(x) = f \circ g(x) = f(g(x))$

$$h'(x) = f'(g(x))\, g'(x)$$

$$\Rightarrow \quad L = \frac{1}{2}(\hat{y} - y)^2 = \frac{1}{2}\left[(w^T x + b) - y\right]^2$$

$$\frac{\partial L}{\partial w} = 2\left(\frac{1}{2}\right)(\underbrace{w^T x + b}_{=2} - \underbrace{y}_{=3})\, x$$

$$= (1)(2 - 3)\, x$$

$$\boxed{\frac{\partial L}{\partial w} = -x = \begin{pmatrix} -2 \\ 1 \end{pmatrix}} \leftarrow \nabla_w L$$

$$\frac{\partial L}{\partial b} = 2\left(\frac{1}{2}\right)(w^T x + b - y)(1)$$

$$\boxed{\frac{\partial L}{\partial b} = (1)(2 - 3)(1) = -1} \leftarrow \nabla_b L$$

c) Using the learning rate $\eta = 0.1$, compute updated parameters W_{new} and b_{new}.

$\Rightarrow$ We use the update rule:

$$\theta_{new} = \theta_{old} - \eta \nabla_\theta L$$

$\Rightarrow$

$$\boxed{W_{new} = W_{old} - \eta \nabla_w L = \begin{pmatrix} 1 \\ 0 \end{pmatrix} - 0.1 \begin{pmatrix} -2 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} - \begin{pmatrix} -0.2 \\ 0.1 \end{pmatrix} = \begin{pmatrix} 1.2 \\ -0.1 \end{pmatrix}}$$

$$\boxed{\Rightarrow \quad b_{new} = b_{old} - \eta \nabla_b L = 0 - 0.1(-1) = 0.1}$$

* Gradient is a vector of partial derivatives

$$f(x,y) \Rightarrow \nabla f = \begin{pmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \end{pmatrix}$$

⇒ The direction of the vector points toward the steepest ascent!

(29) A campaign response model uses 4 inputs, 2 hidden RELU layers, and a 3-class softmax output

a) Given

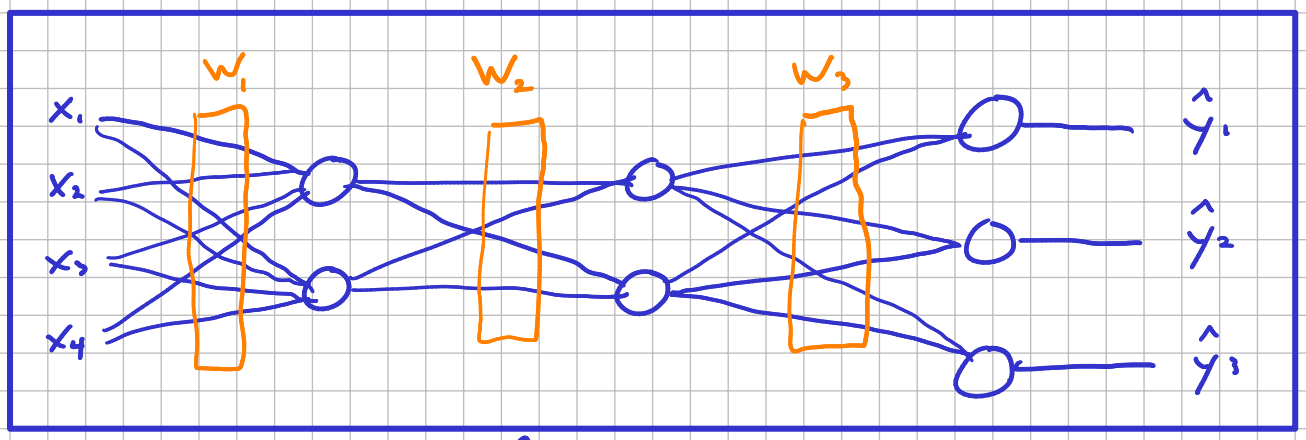
$$x = \begin{pmatrix} 1 \\ 0 \\ -1 \\ 2 \end{pmatrix}, \quad W_1 = \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & -1 & 0 \end{pmatrix}, \quad b_1 = \begin{pmatrix} 0 \\ 0 \end{pmatrix},$$

$$W_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad b_2 = \begin{pmatrix} -2 \\ 0 \end{pmatrix}, \quad W_3 = \begin{pmatrix} 1 & -1 \\ -1 & 1 \\ 0 & 0 \end{pmatrix}, \quad b_3 = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$y = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

Compute forward pass:

- $z^{(1)}, a^{(1)}, z^{(2)}, a^{(2)}, z^{(3)}, \hat{y}$



Architecture

* Note that the number of neurons in each layer matches the number of rows in its weight matrix!

Layer 1

$$z^{(1)} = W_1 x + b_1$$

$$= \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & -1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ -1 \\ 2 \end{pmatrix} = \begin{pmatrix} 1 \cdot 1 + 0 \cdot 0 + 0(-1) + 1 \cdot 2 \\ 0 \cdot 1 + 1 \cdot 0 + (-1)(-1) + 2 \cdot 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$z^{(1)} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$$

$$a^{(1)} = RELU(z^{(1)}) = \max(0, z^{(1)}) = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$$

Layer 2

$$z^{(2)} = W_2 a^{(1)} + b_2$$

$$= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 3 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \end{pmatrix} + \begin{pmatrix} -2 \\ 0 \end{pmatrix}$$

$$z^{(2)} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \qquad a^{(2)} = RELU(z^{(2)}) = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

Layer 3

$$z^{(3)} = W_3 a^{(2)} + b_3$$

$$= \begin{pmatrix} 1 & -1 \\ -1 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$z^{(3)} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \qquad \hat{y} = \text{Softmax}(z^{(3)})$$

$$\text{Softmax} = \frac{e^{z_i}}{\sum e^{z_j}}$$

$$\Rightarrow z_1^3 = 0$$

$$z_2^3 = 0$$

$$z_3^3 = 0$$

$$\hat{y} = \begin{pmatrix} e^0/3e^0 \\ e^0/3e^0 \\ e^0/3e^0 \end{pmatrix} = \begin{pmatrix} 1/3 \\ 1/3 \\ 1/3 \end{pmatrix} \approx \begin{pmatrix} 0.333 \\ 0.333 \\ 0.333 \end{pmatrix}$$

Strategy - Forward Pass and Backprop

1. Forward Pass: Compute pre-activations (z) and the activations (a).

$$z^{(l)} = W_l a^{(l-1)} + b_l$$

2. Errors: Compute the blame (error) vector δ for each level, routing through the transpose of the weight matrix, and activate by the derivative of the activation function.

$$\delta^l = \left(W_{(l+1)}^T \delta^{(l+1)}\right) \odot a^{(l)\prime}$$

$\odot$ Hadamard Product = element-wise multiply

$$\begin{pmatrix} a \\ b \end{pmatrix} \odot \begin{pmatrix} c \\ d \end{pmatrix} = \begin{pmatrix} ac \\ bd \end{pmatrix}$$

3. Gradients: Multiply each layer's blame (error) vector δ by the incoming signal transpose (a^T or x^T) to get the weight/bias gradients

$$\nabla_{W_l} L = \delta^l \left(a^{(l-1)}\right)^T$$

$$\nabla_{b_l} L = \delta^l$$

4. Update the weights and biases

$$\theta_{new} = \theta_{old} - \eta \cdot \nabla_\theta L$$

b) Use cross-entropy with softmax to compute $\delta^{(3)} = \hat{y} - y$, $\delta^{(2)}$, $\delta^{(1)}$

$$\delta^{(3)} = \begin{pmatrix} 0.333 \\ 0.333 \\ 0.333 \end{pmatrix} - \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0.333 \\ -0.667 \\ 0.333 \end{pmatrix}$$

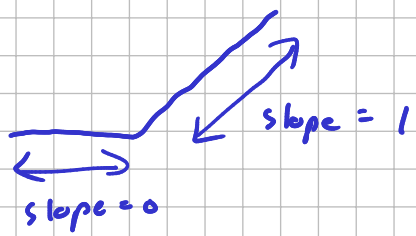
$$\delta^{(2)} = (W_3^T \delta^{(3)}) \odot RELU'(z^{(2)})$$

$$= \begin{pmatrix} 1 & -1 & 0 \\ -1 & 1 & 0 \end{pmatrix} \begin{pmatrix} 0.333 \\ -0.667 \\ 0.333 \end{pmatrix} \odot RELU'(z^{(2)})$$

$$= \begin{pmatrix} 0.33 + 0.67 + 0 \\ -0.33 - 0.67 + 0 \end{pmatrix} \odot RELU'(z^{(2)})$$

$$= \begin{pmatrix} 1.0 \\ -1.0 \end{pmatrix} \odot RELU' \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

* $RELU'(\leq 0) = 0$
  $RELU'(>0) = 1$



$$\delta^{(2)} = \begin{pmatrix} 1.0 \\ -1.0 \end{pmatrix} \odot \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1.0 \\ -1.0 \end{pmatrix} \checkmark$$

$$\delta^{(1)} = (W_2^T \delta^{(2)}) \odot RELU'(z^{(1)})$$

$$= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1.0 \\ -1.0 \end{pmatrix} \odot RELU' \begin{pmatrix} 3 \\ 1 \end{pmatrix} = \begin{pmatrix} 1.0 \\ -1.0 \end{pmatrix} \odot \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$\delta^{(1)} = \begin{pmatrix} 1.000 \\ -1.000 \end{pmatrix}$$

c) Compute all gradients and update:

$W_3, b_3, W_2, b_2, W_1, b_1$

with $\eta = 0.1$

$\Rightarrow$ We use gradient calculations

$$\nabla_{W_2} L = \delta^{(2)} (a^{(2-1)})^T$$

$$\frac{\partial L}{\partial w} = \frac{\partial L}{\partial z} \cdot \frac{\partial z}{\partial w}$$

$$\nabla_{b_2} L = \delta^{(2)}$$

$$\Rightarrow \nabla_{W_1} L = \delta^{(1)} (a^{(1)})^T$$

$$= \begin{pmatrix} 0.333 \\ -0.667 \\ 0.333 \end{pmatrix} \begin{pmatrix} 1 & 1 \end{pmatrix}$$

$$\nabla_{W_1} L = \begin{pmatrix} 0.333 & 0.333 \\ -0.667 & -0.667 \\ 0.333 & 0.333 \end{pmatrix} \checkmark$$

$$\nabla_{b_3} L = \delta^{(3)} = \begin{pmatrix} 0.333 \\ -0.667 \\ 0.333 \end{pmatrix} \checkmark$$